ProofBench compared with Lean benchmarks

Statement statistics use one rule for every benchmark: Lean tokens of the theorem signature, and of the signature with every non-Mathlib definition it uses (same file or the benchmark's own library, followed transitively). Measured on source text, without elaboration.

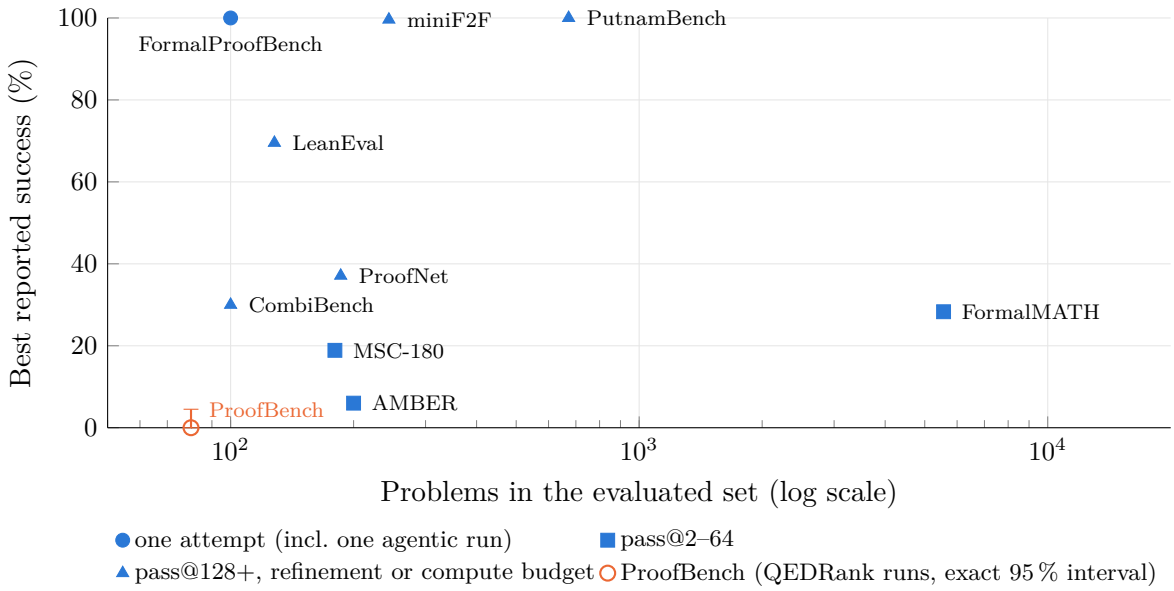


Figure 1: Headroom against size. Each benchmark at its best verified result, placed at the size of the set that result was reported on; marker shape gives the budget behind the number. ProofBench is shown with its exact 95 % Clopper–Pearson interval from the QEDRank runs.

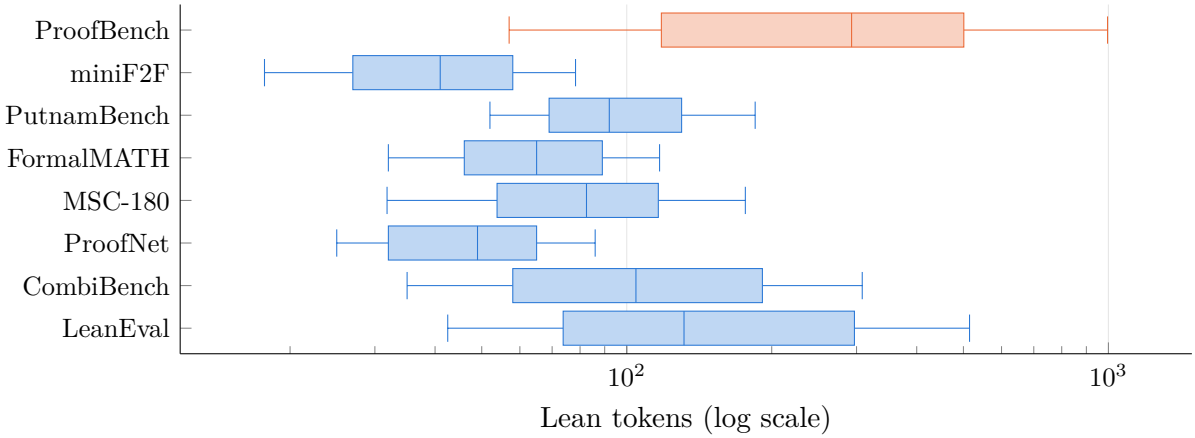


Figure 2: Statement length including the definitions the theorem uses. Box: middle half of the problems; bar: median; whiskers: 10th to 90th percentile.

Table 1: Lean benchmarks for formal mathematics, as described by their authors. ProofBench as of commit `466b32a`.

Benchmark	Problems	Proof assistant	Level	Problems from	Task	Judged by	Statements
ProofBench	111	Lean 4	Research level, including open problems	Recent research papers (mostly graph theory)	Prove the given statement (one sorry); disproof variants	Kernel replay; statement and challenge definitions unchanged; axioms propext, Classical.choice, Quot.sound only	private
miniF2F [zheng2022]	488	Lean (3, Lean 4 ports), Isabelle, HOL Light, Metamath	High-school competition and coursework	AMC, AIME, IMO, the MATH dataset, custom high-school and undergraduate problems	Prove the given statement (answers written into the statement)	No official protocol; each paper checks compilation itself	public
PutnamBench [tsoukalas2024]	672	Lean 4, Isabelle, Coq/Rocq	Undergraduate competition	Putnam competition, 1962–2025	Prove the given statement; about 60% need an exhibited answer (with or without the answer supplied)	Compiles against the repository statements; results need a preprint	public
FormalMATH [yu2025]	5,560	Lean 4	High-school olympiad to undergraduate	Omni-MATH, Numina-Olympiad, AIME, BlueMO, U-Math, HARDMath, DEMIMATH; autoformalized, filtered and expert-reviewed	Prove the given statement (answers embedded)	Lean 4 compiler (leanprover-community/repl)	public
MSC-180 [li2025]	180	Lean 4	Undergraduate to graduate (graduate-dominant)	Undergraduate and graduate textbooks; autoformalized, then rewritten by hand	Prove the given statement	Compilation (kimina-lean-server)	public
ProofNet [azerbayev2023]	371	Lean 3 (Lean 4 ports)	Undergraduate	Textbook exercises (Rudin, Pugh, Stein–Shakarchi, Axler, Dummit–Foote, Herstein, Artin, Munkres, Ireland–Rosen) and Putnam	Theorem proving and (auto)formalization	No proving protocol defined in the paper	public
CombiBench [liu2025]	100	Lean 4	Middle school to IMO and university	Hackmath, Brualdi's textbook, IMO and other olympiads	Prove; construct the answer first for 45 problems (Fine-Eval)	Fine-Eval: compiles, no sorry, no new axioms, statement unchanged, answer matches	public
AMBER [yang2026]	200	Lean 4	Course exercises (easy) to PhD qualifying exams (hard)	Textbooks (Boyd–Vandenberghe, Golub–Van Loan, Vershynin)	Construct-then-prove (82 tasks) and prove (118)	Definition and proof compile within k samples	no public release found
FormalProofBench [ravi2026]	200	Lean 4	Advanced undergraduate and graduate	Qualifying exams and textbooks, formalized by Lean experts	Prove the given statement (agentic, up to 40 turns)	Kernel accepts; #print axioms, no sorryAx; native_decide accepted	private (samples public)
LeanEval	128	Lean 4	Hard, research-level formalization (mostly not open problems)	Community-submitted, reviewed statements; Imperial College Annals Challenge corpus	Prove trusted statements; some problems also ask for definitions or instances	comparator: same statement, permitted axioms only, kernel check	public

Table 2: Best reported results and statement complexity. Statement measures are computed here with one rule for every benchmark.

	Best reported[a]			Statement[b]		
Benchmark	Success	Model	Budget	Signature	With definitions [IQR]	Own definitions
ProofBench	0/80 (0.0 %)	Qwen3 30B-A3B Instruct	pass@1	43	293 [118, 501]	95/111 (86 %)
miniF2F	99.6 %	Seed-Prover	refinement, pass@8–16	41	41 [27, 58]	0/488 (0 %)
PutnamBench	100.0 %	Aleph Prover and 3 others	avg $74 per problem	87	92 [69, 130]	9/672 (1 %)
FormalMATH	28.3 %	DeepSeek-Prover-V2-671B	pass@32	62	65 [46, 89]	626/5558 (11 %)
MSC-180	18.9 %	DeepSeek-Prover-V2-7B	pass@32	63	83 [54, 116]	41/180 (23 %)
ProofNet	37.1 %	DeepSeek-Prover-V2-671B	pass@1024	48	49 [32, 65]	12/371 (3 %)
CombiBench	30.0 %	Seed-Prover	refinement	41	105 [58, 191]	57/100 (57 %)
AMBER	6.0 %	DeepSeek-V3.2 / Gemini-3.0 Pro	pass@16		not measured	
FormalProofBench	100.0 %	4 systems (tie)	one agentic run		not measured	
LeanEval	69.5 %	Axiom Prover	open submissions	49	132 [74, 297]	88/128 (69 %)

[a] As reported by the benchmark's paper or leaderboard; rates are comparable only at equal sampling budget. ProofBench: QEDRank runs, verified answers over established main theorems with a determinate outcome.

[b] Median Lean tokens of the theorem signature, and of the signature together with every non-Mathlib definition it uses (same file or the benchmark's own library, transitively); share of problems whose statement uses at least one such definition.

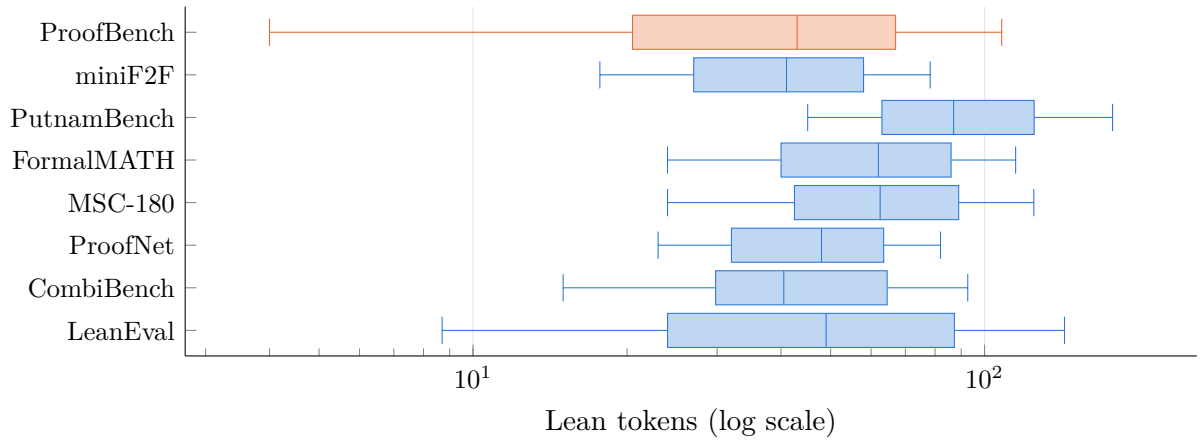


Figure 3: Length of the theorem signature alone. Box: middle half of the problems; bar: median; whiskers: 10th to 90th percentile.

Table 3: Subject coverage: share of labelled problems (%) by subject area; a problem with several labels counts in each. Logic: logic and foundations; Combinatorics: combinatorics and discrete mathematics; Analysis: analysis and calculus; Geometry/topology: geometry and topology; Probability: probability and statistics; Applied: applied mathematics, physics and optimization. *: counts stated in the benchmark's paper.

Benchmark	Logic	Combinatorics	Number theory	Algebra	Analysis	Geometry/topology	Probability	Applied	Other	n
ProofBench	1	87	5	3	0	4	1	0	0	110
miniF2F	0	0	43	57	0	0	0	0	0	313
PutnamBench	1	4	13	42	30	8	1	0	0	672
FormalMATH	0	8	20	47	11	5	0	8	<1	5560
MSC-180	2	3	2	20	32	12	3	27	0	180
ProofNet	0	0	6	53	25	17	0	0	0	359
CombiBench	0	100	0	0	0	0	0	0	0	100
AMBER*	0	0	0	0	25	0	25	50	0	200
FormalProofBench*	11	0	12	20	41	0	17	0	0	200
LeanEval	2	8	28	16	13	32	0	1	1	128

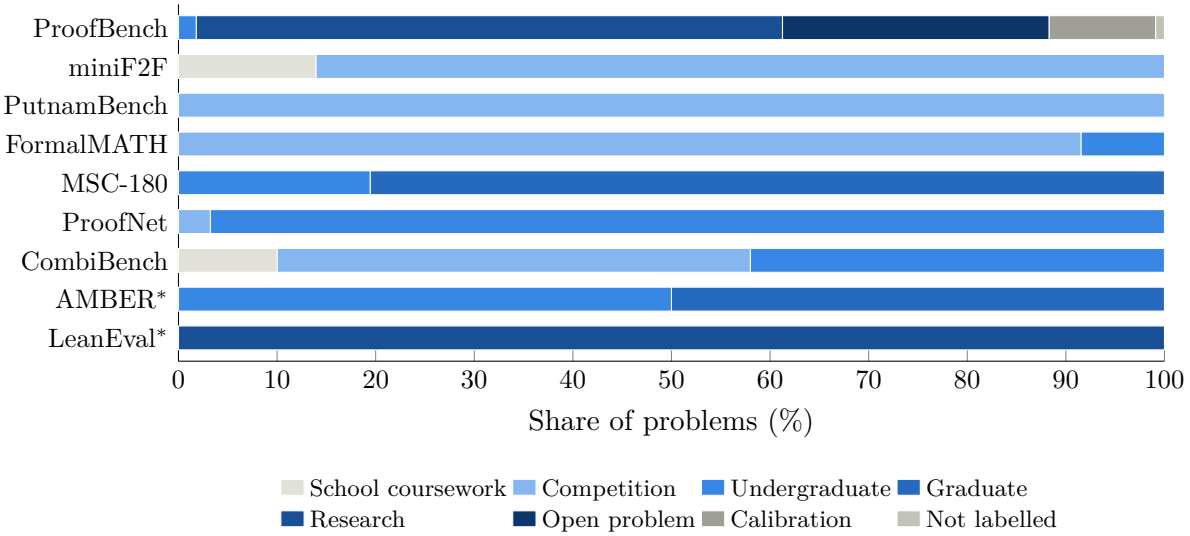


Figure 4: Problems by level. Per-problem labels where the benchmark has them; *: level or counts stated by the benchmark's authors. No level breakdown published: FormalProofBench.

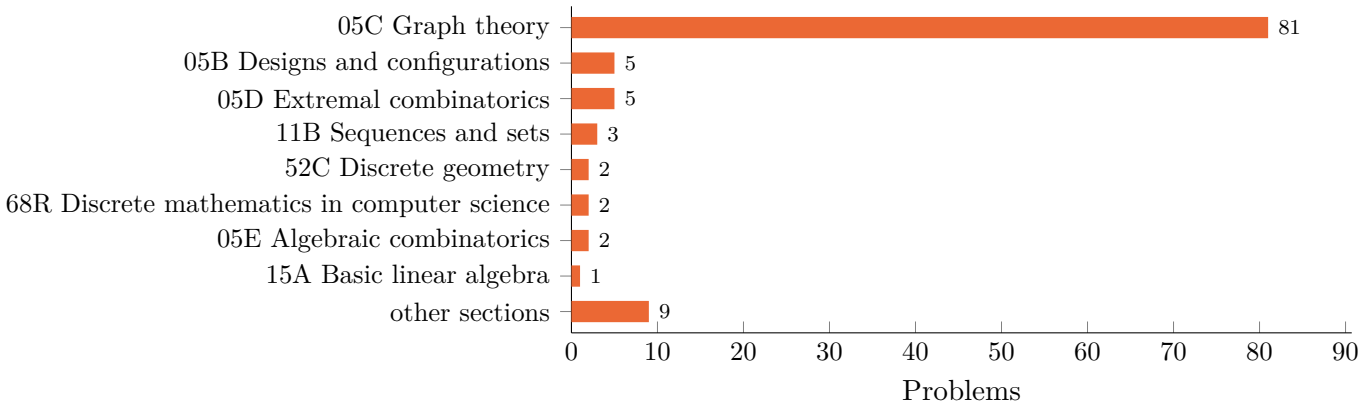


Figure 5: ProofBench problems by MSC 2020 section of their primary subject code (111 problems, 1 without a code).